

NITHEESH PRAHALATH C S

Backend & Edge AI Engineer

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PROFESSIONAL SUMMARY

Backend & Edge AI Engineer with 1.5+ years of experience building production-grade real-time traffic analytics and computer vision systems. Specialized in FastAPI, OpenCV, YOLOv7, TensorRT, and NVIDIA Jetson edge devices. Experienced in RTSP stream processing, multi-threaded Python architectures, radar integration, Linux systems, and AWS cloud pipelines. Proven ability to deliver end-to-end systems from edge inference to cloud event synchronization in high-constraint real-world environments.

TECHNICAL SKILLS

Languages	Python, JavaScript, SQL, Bash
Backend & APIs	FastAPI, Django, REST APIs, WebSockets, JWT Auth, Pydantic
AI / CV	YOLOv7, OpenCV, TensorRT, ONNX, PyTorch, Model Inferencing
Edge & Systems	NVIDIA Jetson Nano / Orin Nano, RTSP Streams, Multi-threading, Linux, FFmpeg, MediaMTX
Frontend	React, Next.js, Electron, Tailwind CSS, Vite
Databases	PostgreSQL, AWS S3
Tools & DevOps	Docker, Git, Bash Scripting, Python Threading / Multiprocessing

PROFESSIONAL EXPERIENCE

DATACORP TRAFFIC PRIVATE LIMITED

Junior Software Engineer (Promoted from Intern)

Bengaluru, India
Jan 2024 – Present

- Developed and maintained real-time traffic violation detection systems processing live RTSP camera feeds on NVIDIA Jetson edge devices using YOLOv7 and TensorRT inference pipelines.
- Configured and managed MediaMTX servers to route and aggregate 3+ simultaneous RTSP camera streams with stable real-time inference and no frame dropouts.
- Built scalable FastAPI and Django backend endpoints for violation event management, system configuration, data transfer, and HTML-based UI integration.
- Implemented multi-threaded Python architectures isolating camera ingestion, AI inferencing, and event generation into independent threads, reducing system processing latency.
- Automated violation asset packaging and cloud synchronization pipelines to AWS S3 for low-latency client recovery and evidence retrieval.

KEY PROJECTS

Car Interceptor & Speed Mapping System | *Company Project* **Stack:** Python, YOLOv7, TensorRT, OpenCV, FastAPI, 4D Radar, NVIDIA Jetson, AWS S3, GPS, PostgreSQL

- Built a mobile edge-based traffic violation detection system processing 3 simultaneous RTSP camera streams on NVIDIA Jetson devices, detecting violations including speeding, no helmet, triple riding, wrong entry, no parking, and lane infractions.
- Integrated 4D radar speed detection modules with YOLOv7 inference output to perform accurate real-time speed mapping to specific moving vehicles with GPS coordinate tracking.
- Engineered a high-throughput multi-threaded pipeline separating camera ingestion, AI inference, and violation event generation into parallel threads using Python threading and structured input queues, minimizing end-to-end latency.
- Personally implemented GPS geo-zone no-parking detection logic, triple-camera feature, cloud data synchronization to AWS S3, data retention policies, and production bug fixes across backend and frontend.
- Violation events generated include plate number, speed, violation type, GPS coordinates, timestamp, captured GIF evidence, and cloud sync status — deployed in production traffic enforcement scenarios.

- Managed deep learning model conversions and optimizations across PyTorch → ONNX → TensorRT for deployment on Jetson edge hardware.

Relay ANPR System – Bengaluru Air Force Base | *Company Project (Lead)* **Stack:** Python, YOLOv7, TensorRT, OpenCV, 4D Radar, PostgreSQL, NVIDIA Jetson

- Led development of a high-security automated vehicle monitoring and ANPR access-control system deployed at the Bengaluru Air Force Base.
- Designed and implemented a custom time + frame buffer mechanism that retains real-time camera frames with timestamps and velocity parameters, triggering inference only when a vehicle exceeds a pre-configured speed threshold from radar input.
- Programmed edge logic to forward buffered frames to the deep learning inference engine at the exact millisecond a speed threshold breach is detected, ensuring precision ANPR capture.
- Adapted existing inference and backend pipeline architecture from the Car Interceptor codebase, customizing for access-control specific logic and security constraints.

Video Downloader — Media Processing Tool | *Personal Project (In Progress)* **Stack:** FastAPI, yt-dlp, FFmpeg, asyncio, PostgreSQL, React, Docker

- Building a backend-heavy media downloader supporting video/MP3 downloads, quality selection, async download queues, FFmpeg transcoding, download history, and retry handling.
- Demonstrates systems thinking with subprocess management, async process handling, file streaming, media pipeline design, and Linux tooling.

Novelex — Story Architecture Desktop App | *Personal Project* **Stack:** React, Electron, Vite, Tailwind CSS, Local JSON Storage

- Architected a standalone desktop environment for novel writing, enabling users to create and manage characters, items, timelines, worlds, and custom categories with cross-entity linking.
- Built a relationship graph canvas rendering visual entity webs and character-to-character relation heatmaps, with full JSON export/import for data portability.

Developer Portfolio & Blog Platform | *Personal Project* **Stack:** Next.js, Vercel, Markdown, JSON

- Built a server-side optimized portfolio using Next.js with a lightweight Markdown-to-JSON parsing pipeline for zero-overhead blog logging and dynamic content delivery.

EDUCATION

SONA COLLEGE OF TECHNOLOGY

Bachelor of Engineering — Computer Science & Engineering | **CGPA:** 8.24 / 10

Salem, Tamil Nadu
Graduated: April 2024

PUBLICATIONS & CERTIFICATIONS

- Fine-Tuning LLMs for Sentiment Analysis — Peer-reviewed research presented at ICCISS'25 International Conference.
- Certified in Blockchain Architecture — Specialized bootcamp by C-DAC Hyderabad & NIELIT Kolkata.
- Global Mentorship Initiative (GMI) — 1-Month Professional Career Preparation graduate track.
- Google-Coursera: Foundations of Cybersecurity | Salesforce Developer Virtual Internship (Nov–Dec 2023) | Anthropic Claude 101.

LANGUAGES

English (Professional Fluency) **Japanese** (Conversational – JLPT N5) **Tamil** (Native) **Hindi / Kannada** (Basic)